

HW 5: DDL June 14, 2026

Python Programming

- 1 Lab # 9 (4 Pts)
- 2 Lab # 10 (4 Pts)

Questions for Tech Report (6 Pts, ≤ 3 Pages)

- 1 Given $\mathbf{X} \in \mathbb{R}^{N \times D}$ and $\mathbf{Y} \in \mathbb{R}^{N \times C}$, where each row of $\mathbf{X}$ is a D -dimensional feature, and each row of $\mathbf{Y}$ is a one-hot vector indicating one of the C classes.
Can we solve classification by a linear regression model? e.g.,

$$\min_{\mathbf{W}} \|\mathbf{Y} - \mathbf{XW}\|_F^2 \quad (22)$$

If it does not work well in general, when it works? (Hint: Find the answer in ESL) (2 Pts)

- 2 Can we solve classification by a nonlinear regression model you learned before? If yes, show your solution. If no, give me your explanation. (2 Pts)
- 3 Derive from (13) to (15) in details. (2 Pts)